

ABSTRACT

ClauseIQ: Machine Learning-Based Contract Clause Classification & Intelligent Search System

Organizations manage a large volume of legal contracts, such as employment and service agreements, where manual retrieval of specific clauses is time-consuming, error-prone, and difficult to scale. To address this, ClauseIQ is proposed as a Machine Learning-based system that automatically extracts clauses from uploaded contracts, classifies them into predefined legal categories, and enables similarity-based search for efficient clause retrieval.

The system is trained and evaluated using the Contract Understanding Atticus Dataset (CUAD), which contains over 13,000 expert-annotated clauses across 41 legal categories. Clause texts are converted into numerical features using Term Frequency-Inverse Document Frequency (TF-IDF). Four classical machine learning classifiers—Logistic Regression, Support Vector Machine (SVM), Naive Bayes, and Random Forest—are trained and compared using metrics like Accuracy and F1-score, while Cosine Similarity ranks clauses by relevance to user queries. Classical machine learning is preferred over deep learning because it is computationally lightweight, interpretable, and appropriate for the project scope.

To provide a user-friendly interface, a web application is developed with a React.js frontend and a FastAPI backend, utilizing Supabase for database management and PyMuPDF and spaCy for text processing. The resulting system automates clause extraction and classification, reduces manual contract review efforts, and delivers fast, relevance-ranked search results to legal professionals.

Dataset Source: Contract Understanding Atticus Dataset (CUAD)

<https://www.atticusprojectai.org/cuad>

Technology Stack:

Frontend: React.js

Backend: FastAPI (Python)

ML Algorithms: Logistic Regression, Support Vector Machine (SVM), Naive Bayes, Random Forest

Database: Supabase (PostgreSQL)

Libraries: PyMuPDF, spaCy

Tools: VS Code, Google Colab, Jupyter Notebook, GitHub

References:

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- [3] M. Kutbi, "Named Entity Recognition Utilized to Enhance Text Classification While Preserving Privacy," *IEEE Access*, vol. 11, pp. 117576–117581, 2023. (<https://doi.org/10.1109/ACCESS.2023.3325895>)

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